SPHERE PACKINGS AND TANGENCY TILINGS IN DIMENSIONS TWO, THREE, AND FOUR

Abstract. Every nonempty open subset of $\mathbb{R}^d$, for $d \in \{2, 3, 4\}$, admits a countably infinite packing of positive-radius closed balls whose common-tangent-halfspace polytopes form a locally finite, face-to-face tiling of exactly that set. Tangencies are precisely facet adjacencies. The numbers of facets are at most 6, 12, and 24, and the outer-to-inner radius ratios are at most 2, $\sqrt{10}$, and $3\sqrt{2} + 2$, respectively. All ball radii can be prescribed to lie below any positive bound. We give the three explicit local constructions using a shared tangency, power-diagram, and stabilization framework. The four-dimensional interpolation step uses an exhaustive exact-integer calculation; its complete checker and the reduction to its finite case list are included.

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1. The theorem and its common framework

Write $B(a, r)$ for the open Euclidean ball and $\overline{B}(a, r)$ for its closure. A polytope is bounded, closed and convex, with nonempty interior. A tiling of an open set $\Omega \subseteq \mathbb{R}^d$ consists of such polytopes contained in Ω, covering Ω, and having pairwise disjoint interiors. It is locally finite relative to Ω if every compact subset of Ω meets finitely many tiles. It is face-to-face if every nonempty intersection of distinct tiles is a face of both, of any codimension. A facet is a face of dimension $d - 1$.

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Theorem 1. *Let $d \in \{2,3,4\}$, let $\Omega \subseteq \mathbb{R}^d$ be nonempty and open, and let $\varepsilon > 0$. There is a countably infinite family $\{\overline{B}(a, r_a) : a \in V\}$ of positive-radius closed balls with pairwise disjoint interiors such that the following holds. Write $a \sim b$ when $|a - b| = r_a + r_b$ and define*

$$H^-_{a,b} = \left\{ x \in \mathbb{R}^d : (x-a) \cdot \frac{b-a}{|b-a|} \le r_a \right\}, \qquad a \sim b, \tag{1.1}$$

$$T_a = \bigcap_{b \sim a} H^-_{a,b}. \tag{1.2}$$

The sets T_a form a locally finite, face-to-face tiling of exactly Ω. Two balls are tangent if and only if their tiles share a facet, and that facet lies in their common tangent hyperplane. Moreover,

$$0 < r_a \le \varepsilon, \qquad \overline{B}(a, r_a) \subseteq T_a \subseteq \overline{B}(a, C_d r_a),$$

and every tile has at most N_d facets, where

d	2	3	4
N_d	6	12	24
C_d	2	$\sqrt{10}$	$3\sqrt{2}+2$

In particular, joining ball centers across shared facets gives an orthogonal tiling graph: the center-to-center segment is perpendicular to that facet.

There is no boundedness, connectedness or boundary-regularity assumption on Ω. Coverage is by the polytopes, not by the balls. Local finiteness is relative to Ω, so accumulation at its boundary is allowed. For example, one may take $\Omega = U \setminus F$ for any open U and closed F with nonempty difference, without a measure or dimension restriction on F. No result for $d \ge 5$ is asserted. The notion of an orthogonal tiling agrees with that in [BG25, Section 1.1].

2. Paired tilings and power comparisons

A *paired tiling* is a locally finite, face-to-face tiling supplied with a positive inball in every tile, such that every facet is tangent to that ball and the two contact points on every common facet coincide.

Lemma 2. *In a paired tiling of an open set, the inballs form a packing. Their tangencies are precisely the facet adjacencies, and every tile is the intersection of its ball's common-tangency halfspaces.*

Proof. A ball's interior lies in the interior of its containing tile, so ball interiors are disjoint. A ball cannot touch a face of codimension at least two of its tile: two distinct supporting facet planes would then support a sphere at the same point, which is impossible.

Suppose two balls touch at p. Then p lies on both tile boundaries. Indeed, if p were interior to one tile, interior points of the other ball arbitrarily near p would belong to both tile interiors. Thus p is in the relative interior of a facet of each tile. The face-to-face property identifies those facets. The converse is exactly the matching-contact condition.

Finally, every tile facet has a neighboring tile. Approach a relative interior point from the other side within a compact ball contained in the open domain. Local finiteness gives one repeatedly encountered tile. Coverage, closedness, and the face-to-face condition show that it shares the entire facet. The common-tangency halfspaces are therefore exactly the facet halfspaces. □

For a ball define its *power* and its affine score by

$$\pi_{a,r}(x) := |x - a|^2 - r^2, \qquad \ell_{a,r}(x) := 2a \cdot x - |a|^2 + r^2. \tag{2.1}$$

Lemma 3. *Every whole-space paired tiling is the power diagram of its balls: the tile at a is the set on which π_{a,r_a} is minimal among all ball powers.*

Proof. Across a facet with unit normal n pointing from a to b, matching contacts give

$$b - a = (r_a + r_b)n. \tag{2.2}$$

The affine scores agree on its plane. At a point p in a lower-dimensional intersection, take a small ball meeting only tiles that contain p. Such a ball exists by local finiteness and closedness. A generic polygonal path connects the interiors of any two incident tiles through facet interiors in this ball, so agreement propagates to p. The affine scores consequently define one continuous, locally piecewise-affine function f.

Across a facet, the gradient jumps from $2a$ to $2b$ in the positive normal direction. Along a generic line segment every slope jump is nonnegative. Such segments avoid the finitely many faces of codimension at least two in a compact neighborhood; approximation and continuity give convexity on every segment. Each affine piece is a global supporting affine function of f, whence

$$f(x) = \max_{a \in V} \ell_{a,r_a}(x). \tag{2.3}$$

Its equality set with a given affine piece is exactly the corresponding tile. Otherwise convexity, together with an interior ball of that tile, would force equality on an open set outside it. Local finiteness would then force two affine functions with different gradients to agree on an open set. Taking $|x|^2 - f(x)$ proves the assertion. □

We use Lemma 3 only for whole-space tilings at finite stages. No assertion of global convexity on a nonconvex set Ω is needed.

Lemma 4 (Simultaneous insertion)**.** *Let $L \subset \mathbb{R}^d$ be a lattice, let $r > 0$, and let $\mathcal{R} \subset L$ be a finite set of vectors of length $2r$. Start from a whole-space paired tiling. Let $A \subset L$ be a set of new centers of radius r, with the enlarged ball family having disjoint interiors. For each $c \in A$, suppose*

$$W_c = \bigcap_{u \in N_c} \{x : u \cdot (x - c) \le 2r^2\}, \qquad N_c \subseteq \mathcal{R},$$

is a polytope contained in $\overline{B}(c, M)$ for a uniform M, and assume:

1. *Each $c + u$, $u \in N_c$, is either another new center or an old center whose radius is r.*
2. *On W_c, the new power $\pi_{c,r}$ is no greater than the old minimum power and is no greater than $\pi_{b,r}$ for every $b \in A$.*

Then the enlarged power diagram is a paired tiling. The new cell of c is exactly W_c, and old cells only shrink. If all W_c lie in the union of a specified subfamily of old tiles, every old tile outside that subfamily remains unchanged.

Proof. The second assumption puts W_c in the new cell. Outside W_c, a violated inequality and the neighbor from the first assumption give strictly smaller power. This identifies the new cell exactly.

The old minimum is attained by Lemma 3. At each fixed point, new powers tend to infinity as their lattice centers tend to infinity, so the enlarged minimum is attained and the cells cover the whole space. Old cells are intersected with new power halfspaces and retain their balls: a point in one of two nonoverlapping balls has nonpositive power for that ball and nonnegative power for the other. On a fixed old polytope only finitely many new inequalities can affect its own power cell. Thus old cells remain bounded full-dimensional polytopes. New cells contain their balls because the defining planes are at distance r from their centers. Old local finiteness, lattice discreteness, and the uniform bound M give local finiteness. Power-bisector inequalities imply that cell intersections are faces of both cells.

Every facet of W_c comes from a root $u \in N_c$. Its points also minimize the power for $c + u$, so that neighbor contains the facet; the two equal balls touch at $c + u/2$. Old–old facets lie in old facet planes. Their contact points survive because the old balls remain in their cells. Hence all shared facets have matching contacts.

If a point is removed from an old cell, some minimizing new site has strictly smaller power, and that point belongs to the union of the new cells. If an unselected old tile lost a boundary

point, continuity of this strict inequality would make it lose interior points as well, which is impossible since its interior is disjoint from the selected union. It therefore remains unchanged as a closed polytope. □

3. A shared stabilization principle

All three constructions below use whole-space paired tilings at finite stages. It is important to retain complete pairs, not disks whose centers merely appeared at some earlier stage.

Lemma 5 (Whole-pair stabilization)**.** *Let $F \subset \mathbb{R}^d$ be nonempty and closed, with $\Omega = \mathbb{R}^d \setminus F$ nonempty. Suppose $(\mathcal{T}_n)_{n\geq 0}$ are whole-space paired tilings and there are sets S_n with the following properties:*

1. *Every complete pair removed or introduced in $\mathcal{T}_n \to \mathcal{T}_{n+1}$ has its tile contained in S_n. For every compact $K \subset \Omega$, all sufficiently late S_n avoid K.*
2. *For every n, a pair of $\mathcal{T}_n$ whose tile meets F is absent from some later $\mathcal{T}_m$, $m > n$.*

Then the eventually unchanged pairs

$$\mathcal{T}_\infty = \bigcup_{N\geq 0} \bigcap_{n\geq N} \mathcal{T}_n$$

form a countably infinite, locally finite paired tiling of exactly Ω. Every radius, facet-count or outer-to-inner-radius bound holding at all finite stages is inherited by the final pairs. Their tiles are their balls' tangency polytopes.

Proof. For compact $K \subset \Omega$, choose N after which all supports avoid K. No subsequent step removes a pair meeting K or introduces a new one meeting K. Hence the full collection of pairs meeting K is the same at stage N and in the limit. It is finite and covers K. Condition (2) excludes every eventual pair meeting F, so coverage is exactly Ω and all limiting tiles are contained in Ω.

Two surviving pairs occur together at a sufficiently late stage. Their balls have disjoint interiors and their tiles meet face-to-face. Matching contacts across a facet also persist; equivalently, they can be checked in a compact neighborhood of a relative interior point of the facet. Every pair inherits its positive-radius inball and all finite-stage bounds. Thus the limit is a paired tiling, and Lemma 2 identifies its tangency polytopes and its tangencies with facet adjacencies.

Every finite-stage family is countable, by local finiteness and exhaustion of the whole space by compact balls. The limit is a subset of the union of these countably many families. It cannot be finite: a finite union of bounded closed polytopes is compact, whereas a nonempty open subset of $\mathbb{R}^d$ is not compact. This proves countable infinitude. □

4. The planar construction

4.1. **The standard hexagonal tiling.** Set

$$a = (2, 0), \quad b = (1, \sqrt{3}), \quad \Lambda = \mathbb{Z}a + \mathbb{Z}b, \quad n_k = (\cos(k\pi/3), \sin(k\pi/3)), \quad 0 \leq k < 6,$$

with indices modulo six. Put

$$H = \{x : n_k \cdot x \leq 1 \text{ for all } k\}, \qquad F_k = H \cap \{x : n_k \cdot x = 1\}.$$

Its vertices are $(1, \pm 1/\sqrt{3})$, $(-1, \pm 1/\sqrt{3})$ and $(0, \pm 2/\sqrt{3})$, so its inradius is 1 and circumradius is $2/\sqrt{3}$. The cells $z + H$, $z \in \Lambda$, form the regular hexagonal paired tiling with disks $\overline{B}(z, 1)$.

Indeed, $|ma + nb|^2 = 4(m^2 + mn + n^2)$. The six shortest nonzero vectors are $2n_k$; all others have length at least $2\sqrt{3}$, because $m^2 + mn + n^2$ is never 2 modulo 3. For $x \in H$ and any such other vector λ,

$$x \cdot \lambda \leq (2/\sqrt{3})|\lambda| \leq |\lambda|^2/2.$$

Thus H is the lattice Voronoi cell, its six inequalities are precisely the nonredundant nearest-point inequalities, and the cells cover the plane face-to-face. Across $z + F_k$, the disks at z

and $z + 2n_k$ touch at $z + n_k$. Write $\Lambda_R = R\Lambda$; the standard radius-R pair at $z \in \Lambda_R$ is $(\overline{B}(z, R), z + RH)$.

Lemma 6 (Planar local refinement)**.** *Let $S \subseteq \Lambda_R$. Suppose a whole-plane paired tiling contains the complete standard radius-R pair at each $z \in S$. These pairs can be replaced while every other old pair remains unchanged. Every new disk has radius $r = R/3$, and its polygon Q with center c has at most six sides and satisfies*

$$\overline{B}(c, r) \subseteq Q \subseteq \overline{B}(c, 2r).$$

Every new polygon lies in $\bigcup_{z\in S}(z + RH)$. If $c \in \Lambda_{R/3}$ and every $z \in \Lambda_R$ with $|z - c| < 3R$ belongs to S, the new tiling contains the complete standard radius-$R/3$ pair at c.

Proof. It suffices to work at $R = 1$.
First operation: subdivide a selected hexagon. Replace H and its disk by a central pair with polygon $H/3$ and center 0, and six pairs with

$$T_k = \operatorname{conv}(F_k \cup (F_k/3)), \qquad c_k = \tfrac{2}{3}n_k.$$

All seven disks have radius $1/3$. The polygons partition H: intersect each ray from the origin first with $H/3$ and then with the corresponding trapezoid. Their intersections are full edges or vertices. For example,

$$T_0 = \{(x, y) : 1/3 \le x \le 1,\ |y| \le x/\sqrt{3}\}. \tag{4.1}$$

Each of its four side lines has distance $1/3$ from $(2/3, 0)$. The central disk is inscribed in $H/3$. The distances between a central and peripheral center, and between consecutive peripheral centers, are $2/3$. If q is the common endpoint of F_k, F_{k+1}, the latter contact is $q/2$, in the relative interior of $[q/3, q]$.

The disk at c_k meets the old side at n_k. An outside disk of radius ρ formerly meeting that side has center $(1 + \rho)n_k$ by matching contacts, so the new disk touches it at the same point. If both old cells are selected, their new centers across that side are $(2/3)n_k$ and $(4/3)n_k$, again with contact n_k. No old boundary edge is subdivided. The vertices $(1, \pm 1/\sqrt{3})$ and $(1/3, \pm 1/(3\sqrt{3}))$ of T_0 are at distances $2/3$ and $2/(3\sqrt{3})$ from its center. Convexity gives the shape bound; the central hexagon satisfies it as well.
Second operation: insert a disk at an eligible old vertex. An old vertex q is eligible if all three incident old hexagons were selected. Its six incident peripheral centers are $q + (2/3)n_k$. Insert the disk at q of radius $1/3$ with polygon $q + H/3$, cutting one corner from each of these six trapezoids. At $q_+ = (1, 1/\sqrt{3})$, the retained part of T_0 is

$$T_0 \cap \{x + \sqrt{3}y \le 4/3\}.$$

The removed closed triangle has vertices

$$(1, 1/\sqrt{3}), \quad (1, 1/(3\sqrt{3})), \quad (2/3, 2/(3\sqrt{3})).$$

The last two are consecutive vertices of $q_+ + H/3$. Rotations by $2\pi/3$ about q_+ and reflection in $x = 1$ give the six center-to-side triangles that partition that hexagon. Each cut is the common tangent line of the retained disk and the inserted disk. The polygons therefore cover exactly the same region. Cuts on both sides of each old shared edge agree, and every new edge is a full side of the inserted hexagon. This proves the face-to-face property.

Each old peripheral disk lies in every retained halfplane, so its old contact points survive. None can become a polygon vertex, by the unique supporting-line argument in Lemma 2. Old shared edges retain positive length and matching contacts; the new shared edges have matching contacts with the inserted disk.

Old vertices form the two cosets $\pm(a + b)/3 + \Lambda$. Within one coset their separation is at least 2. Between the two cosets, squared distances have the form

$$\tfrac{4}{9}(u^2 + uv + v^2), \qquad u, v \in \mathbb{Z}, \quad u \equiv v \equiv 2 \pmod 3.$$

The parenthesis is a positive multiple of 3, so distinct old vertices are at distance at least $2/\sqrt{3}$, greater than twice the inserted polygon's circumradius $2/(3\sqrt{3})$. Thus inserted polygons are disjoint and the cuts commute. Each lies in its three selected old cells. A selected–unselected edge has no eligible endpoint and is not subdivided by these cuts. Unselected old pairs are unchanged even on their boundaries. Each trapezoid loses at most its two outer corners, so all new polygons have at most six sides. Truncation preserves both the inball and the outer-radius bound. If both outer vertices of T_0 are eligible, its resulting cell is

$$T_0 \cap \{x + \sqrt{3}y \le 4/3\} \cap \{x - \sqrt{3}y \le 4/3\} = (2/3, 0) + H/3. \tag{4.2}$$

The four inequalities in (4.1) and these two cuts are precisely the six side inequalities on the right.

Complete fine cells and local finiteness. The nine classes of $\Lambda/3$ modulo Λ have representatives

$$0, \quad \pm a/3, \quad \pm b/3, \quad \pm(a-b)/3, \quad \pm(a+b)/3.$$

The middle six are the uniquely represented peripheral points $z + (2/3)n_k$, and the last two are old vertices. An old center needs only its own cell selected. An old vertex needs its three incident cells, whose centers are at distance $2/\sqrt{3} < 3$. For a peripheral point $c = z + (2/3)n_k$, it suffices to select

$$z, \quad z + 2n_k, \quad z + 2n_{k-1}, \quad z + 2n_{k+1}.$$

These centers are incident to either endpoint of $z+F_k$ and all lie within distance $2+2/3 < 3$ of c. Both endpoints are eligible, so (4.2) gives the complete smaller cell. Further selections cannot alter these complete pairs: central hexagons are never cut, inserted hexagons are disjoint, and a peripheral trapezoid has only the two specified possible cuts. Each new polygon is in the selected old cells incident to it, with only finitely many new polygons associated to one old cell. A compact set meets finitely many old cells and hence finitely many new polygons. □

4.2. **Refinement near the complement.**

Proof of Theorem 1 in dimension two. Put $F = \mathbb{R}^2 \setminus \Omega$. If F is empty, use the standard radius-ε tiling. Otherwise set $R_n = \varepsilon 3^{-n}$ and $\Lambda_n = R_n\Lambda$. Construct whole-plane paired tilings $\mathcal{T}_n$ with radii in $[R_n, \varepsilon]$, at most six sides per tile, and shape ratio at most 2, maintaining

$$z \in \Lambda_n, \quad \operatorname{dist}(z, F) \le 12R_n \quad \Longrightarrow \quad (\overline{B}(z, R_n), z + R_nH) \in \mathcal{T}_n. \tag{4.3}$$

Start with the standard scale R_0. At stage n, apply Lemma 6 to $S_n^* = \{z \in \Lambda_n : \operatorname{dist}(z, F) \le 8R_n\}$. These complete pairs are present by (4.3). If $\operatorname{dist}(c, F) \le 12R_{n+1}$ and $|z - c| < 3R_n$, then

$$\operatorname{dist}(z, F) < (4 + 3)R_n < 8R_n.$$

The complete-cell assertion proves (4.3) at the next stage. Every removed or introduced polygon lies in

$$\{x : \operatorname{dist}(x, F) < 10R_n\},$$

because selected centers have distance at most $8R_n$ and their standard cells have circumradius $2R_n/\sqrt{3} < 2R_n$. Unselected complete pairs remain unchanged. These supports avoid every compact subset of Ω for all sufficiently large n.

Every tile meeting F is removed at the next step. To see this, let $x \in Q \cap F$ for a tile at stage n. Every standard radius-R_n hexagon meeting $B(x, R_n)$ has center within distance $R_n + 2R_n/\sqrt{3} < 3R_n$ of x, so its complete pair occurs by (4.3). These finitely many hexagons cover $B(x, R_n)$. The set $\operatorname{int} Q \cap B(x, R_n)$ is nonempty and open, even if x is a vertex. Finitely many standard-cell edges cannot cover it. Thus the interior of Q meets a standard-cell interior; disjointness identifies the complete pairs. Its center has distance at most $2R_n/\sqrt{3} < 8R_n$ from F, so this pair is selected and replaced by smaller disks. Lemma 5 now gives the desired tiling and all its bounds. □

5. The three-dimensional construction

5.1. **The lattice and two elementary polytopes.** In this section set

$$L = \{z \in \mathbb{Z}^3 : z_1 + z_2 + z_3 \text{ is even}\}, \qquad \mathcal{R} = \{\pm e_i \pm e_j : 1 \le i < j \le 3\}, \qquad r_* = 1/\sqrt{2},$$

where the signs are independent, and put

$$P = \{x : u \cdot x \le 1 \text{ for } u \in \mathcal{R}\} = \{x : |x_i| + |x_j| \le 1 \text{ for } i < j\}.$$

Lemma 7. *The polytope P is the Voronoi cell of L at the origin. Its vertices are*

$$\pm e_i \ (1 \le i \le 3), \qquad (\pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}),$$

and $\overline{B}(0, r_*) \subseteq P \subseteq \overline{B}(0, 1)$. For every $h > 0$ the pairs $(\overline{B}(z, hr_*), z + hP)$, $z \in hL$, form a paired tiling of $\mathbb{R}^3$.

Proof. The inequalities give the vertex list. At $x = \pm e_i$, the Voronoi inequality $2x \cdot z \le |z|^2$ follows from $2|z_i| \le |z|^2$ for $z \in L$: if $|z_i| \ge 2$, use $2|z_i| \le z_i^2$, whereas $|z_i| = 1$ forces another nonzero coordinate by parity. At a half-sign vertex,

$$2x \cdot z \le \sum_i |z_i| \le \sum_i z_i^2.$$

Thus every lattice comparison holds on P, and the root comparisons among them already define P. The facet distances and vertex list give the ball inclusions. Across the facet indexed by $u \in \mathcal{R}$, the balls at 0 and u meet at $u/2$. Translation and scaling finish. □

For $i \in \{1, 2, 3\}$ let

$$U_i = \{u \in \mathcal{R} : u_i = 0\}, \qquad K_i = \{y : |y_i| + |y_j| \le 1 \text{ for } j \ne i\}.$$

Lemma 8. *The vertices of K_i are $\pm e_i$ and the four vectors $\pm e_j \pm e_k$, where $\{i, j, k\} = \{1, 2, 3\}$. Moreover,*

$$K_i = \{y : 2y \cdot v \le |v|^2 \text{ for every } v \in L \setminus U_i\}. \tag{5.1}$$

It is circumscribed about $\overline{B}(0, r_*)$ and contained in $\overline{B}(0, \sqrt{2})$.

Proof. At fixed y_i, the other coordinates range independently over $[-(1 - |y_i|), 1 - |y_i|]$, giving the vertices. At $\pm e_i$ the comparisons in (5.1) follow as above. At a vertex $y \in U_i$, every $v \in L \setminus U_i$ satisfies $|v - y|^2 \ge 2$, because $v - y$ is a nonzero lattice vector. Expanding proves the comparison. Conversely, the constraints indexed by $\mathcal{R} \setminus U_i$ define K_i. Their planes have distance r_* from the origin, and the vertices have norm at most $\sqrt{2}$. □

5.2. **Three local operations.** Call $(\overline{B}(z, hr_*), z + hP)$ a standard scale-h pair.

Proposition 9 (Three-dimensional local refinement)**.** *Suppose a whole-space paired tiling contains the complete standard scale-h pair at every point of $S \subseteq hL$. These pairs may be replaced, with all other old pairs unchanged, so that:*

1. *every new ball has radius $hr_*/3$, and its tile lies in $\bigcup_{z \in S}(z + hP)$;*
2. *every new tile has at most twelve facets and outer-to-inner radius ratio at most $\sqrt{10}$;*
3. *if $c \in (h/3)L$ and every $z \in hL$ with $|z - c| \le 10h$ belongs to S, the complete standard scale-$h/3$ pair at c is present.*

Proof. Scale to $h = 3$, so $S \subseteq 3L$ and every new radius is r_*.
First operation: subdivide selected cells. For the selected parent $3P$ at 0, take the central pair $(\overline{B}(0, r_*), P)$ and twelve balls at $u \in \mathcal{R}$, with cells

$$C_u = \{x \in 3P : u \cdot x \ge 1, \ u \cdot x \ge v \cdot x \text{ for all } v \in \mathcal{R}\}. \tag{5.2}$$

For $u = e_1 + e_2$ this is

$$C_u = \{x : 1 \le x_1 + x_2 \le 3, \ x_1 \ge |x_3|, \ x_2 \ge |x_3|\}. \tag{5.3}$$

Other cases follow by signed permutations. Outside P, choose a root maximizing $u \cdot x$, proving coverage. These are precisely the equal-radius Voronoi cells of $\{0\} \cup \mathcal{R}$ clipped to $3P$, so they

meet face-to-face. All six planes in (5.3) have distance r_* from $(1,1,0)$. The inner neighbor is the central ball. Across $x_1 = x_3$, for example, the neighbor center is $(0,1,1)$ and the difference is a root; signed permutations give all internal contacts.

At the outer facet $u \cdot x = 3$, the new ball touches at $3u/2$, the old ball's contact point. That whole facet is unchanged, and its intersections with other children are its old lower-dimensional faces. Thus the induced boundary subdivision is unchanged and the replacements fit unselected neighbors and each other. The first operation gives a paired tiling. The vertices of (5.3) are

$$(1,0,0), \quad (0,1,0), \quad (\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}), \quad (\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2})$$

and their triples. Their distances from $(1,1,0)$ are at most $\sqrt{5} = \sqrt{10}r_*$. The central cell satisfies a stronger bound. After translating to all selected parents, every cell inside that region has center in L and radius r_*; outside it, old complete pairs remain.

Second operation: insert at six-cell vertices. Set $Q = \{3m : m \in \mathbb{Z}^3,\ m_1 + m_2 + m_3 \text{ is odd}\}$. Call $q \in Q$ eligible if all six centers $q \pm 3e_i$ belong to S. For each eligible q, insert the six balls at

$$c = q + \sigma e_i, \qquad 1 \le i \le 3, \quad \sigma \in \{-1,1\},$$

with proposed cells $W_c = c + K_i$. We check Lemma 4. The cell W_c lies in the selected parent centered at $z = q + 3\sigma e_i$: relative to z, its vertices are $-3\sigma e_i$, $-\sigma e_i$ and $-2\sigma e_i \pm e_j \pm e_k$, all in $3P$.

The new centers lie in L by parity. Their single nonzero residue modulo 3 determines i, σ, q uniquely. They are distinct and were not produced by the first operation, whose centers have zero or two nonzero residues. The minimum nonzero lattice length is $2r_*$, so the new balls and earlier equal-radius balls have disjoint interiors. Containment in selected parents gives disjointness from unselected old ball interiors.

Current tiles covering this parent have centers $b = z$ or $z + v$, $v \in \mathcal{R}$. The ith coordinate of $b - c$ is among $\sigma, 2\sigma, 3\sigma$, hence $b - c \notin U_i$. Lemma 8 gives $\pi_{c,r_*} \le \pi_{b,r_*}$ on W_c. At each point choose a current tile in that parent. Its power is the current minimum by Lemma 3, proving comparison with every current ball. If c' is another insertion, then $c' - c \notin U_i$: adding a vector of U_i to c produces three nonzero residues, while c' has one. Lemma 8 therefore also gives all simultaneous comparisons.

The eight facet normals of K_i are roots with nonzero ith coordinate. Their required neighbors are, for $j \ne i$ and $\tau \in \{-1,1\}$,

$$c - \sigma e_i + \tau e_j = q + \tau e_j, \qquad c + \sigma e_i + \tau e_j = z - \sigma e_i + \tau e_j.$$

The first is an insertion at the same q, the second a first-operation site in this parent. Lemma 4 therefore produces exactly the proposed cells, with circumradius at most $\sqrt{2}$, shrinks earlier selected cells and leaves unselected pairs unchanged.

Third operation: insert four-point clusters. For $b \in L$ let $k(b)$ count its nonzero residues modulo 3. If all coarse parents were selected, the first operation would give exactly $k = 0, 2$, and the second exactly $k = 1$. For $k = 2$, representing each nonzero residue by 1 or -1 gives uniquely $b = z + u$, $u \in \mathcal{R}$, and parity gives $z \in 3L$. For $k = 1$, write uniquely $b = q + \sigma e_i$ with $q \in Q$. The remaining sites form disjoint clusters

$$T_m = (3m + \{1,2\}^3) \cap L, \qquad q_m = 3m + (\tfrac{3}{2},\tfrac{3}{2},\tfrac{3}{2}), \qquad m \in \mathbb{Z}^3.$$

Each has four sites, any two differing by a root. If $c \in T_m$ and $u \in \mathcal{R}$, then $c + u$ either has $k \le 2$ or belongs to the same cluster: preserving nonzero residues forces the two changed coordinates to switch between $3m_j + 1$ and $3m_j + 2$.

Call m eligible if $\{z \in 3L : |z - q_m| \le 12\} \subseteq S$. Insert all four sites of every eligible cluster, with cells $W_c = c + P$. Residues show distinctness from earlier sites and between clusters. Since $|c - q_m| = \sqrt{3}/2$ and $P \subseteq \overline{B}(0,1)$, any coarse cell meeting $c + P$ has center within distance $\sqrt{3}/2 + 1 + 3 < 5$ of q_m and was selected. Thus current equal-radius lattice cells cover $c + P$. The lattice Voronoi property proves comparison with their minimum power and with every other new site. Containment and lattice separation give nonoverlap.

To supply a root neighbor $b = c + u$, if $k(b) = 3$ it is in the same cluster. Otherwise $|b - q_m| \le \sqrt{3}/2 + \sqrt{2}$. For $k = 0, 2$, its first-operation parent is within a further $\sqrt{2}$. For $k = 1$, write $b = q' + \sigma e_i$; all six parents needed to make q' eligible are within a further $1+3$ of b. Every required parent is therefore within 12 of q_m and selected. Lemma 4 supplies precisely $c + P$ and preserves unselected pairs.

Bounds and complete fine cells. All new facet normals are roots, and later insertions only cut earlier new cells by root-normal planes. The twelve oriented roots give at most twelve facets. The circumradius bounds are $\sqrt{5}$, $\sqrt{2}$, and 1 in the three operations; taking power cells only shrinks earlier cells. All changes stay in the selected coarse region, and local finiteness follows from Lemma 4 and the first subdivision.

A site $b \in L$ is produced if every coarse center within 13 of b is selected: the required radii are at most $\sqrt{2}$, 4, and $12 + \sqrt{3}/2 < 13$ for the three operations. Selecting all coarse centers within 30 of c therefore produces c and its twelve root neighbors, since $13 + \sqrt{2} < 30$. Those neighbor comparisons bound its cell by $c + P$. Conversely every coarse cell meeting $c + P$ has center within 4 of c and is selected. A current equal-radius lattice cell attains the minimum at every point of $c + P$, and the full lattice inequalities show that c has no greater power. Its cell is exactly $c + P$. Rescaling from $h = 3$ proves the proposition. □

5.3. **Refinement near the complement.**

Proof of Theorem 1 in dimension three. Let $F = \mathbb{R}^3 \setminus \Omega$. If it is empty, use the standard tiling at scale $\sqrt{2}\varepsilon$. Otherwise put $h_n = \sqrt{2}\varepsilon 3^{-n}$ and $L_n = h_n L$. Starting with the standard scale-h_0 tiling, maintain

$$z \in L_n, \quad \operatorname{dist}(z, F) \le 30h_n \quad \Longrightarrow \quad (\overline{B}(z, h_n r_*), z + h_n P) \in \mathcal{T}_n.$$

Apply Proposition 9 to $S_n^* = \{z \in L_n : \operatorname{dist}(z, F) \le 20h_n\}$. If $\operatorname{dist}(c, F) \le 30h_{n+1} = 10h_n$ and $|z - c| \le 10h_n$, then $\operatorname{dist}(z, F) \le 20h_n$, proving the next complete-pair invariant. Every removed or introduced tile is in

$$\{x : \operatorname{dist}(x, F) \le 21h_n\},$$

because standard scale-h_n cells have circumradius h_n and every new tile lies in the selected old union. Radii lie in $[h_n r_*, \varepsilon]$, with the stated facet and shape bounds.

If a tile Q meets F at x, the standard cells meeting $B(x, h_n)$ have centers within $2h_n$ of x, so the invariant provides their complete pairs. They finitely cover that ball. The nonempty open set $\operatorname{int} Q \cap B(x, h_n)$ cannot lie in their finitely many boundaries. It meets a standard-cell interior, identifying Q and its complete pair with that standard pair. Its center is within h_n of F and is selected for replacement. Lemma 5 now proves the theorem. □

Tangency defect. For a polytope Q containing $\overline{B}(v, r)$, let h_G be the distance from v to a facet plane. Decomposition into pyramids gives

$$3|Q| - r\mathcal{H}^2(\partial Q) = \sum_{G \text{ facet of } Q} \mathcal{H}^2(G)(h_G - r) \ge 0.$$

The defect vanishes exactly when Q is circumscribed about the ball. Every intermediate pair above has zero defect; no improvement assertion for a positive-defect energy minimizer is used.

6. The four-dimensional construction

In this section all lattices, roots, and polytopes lie in $\mathbb{R}^4$.

6.1. **The D_4 lattice and a subdivision preserving whole facets.** Let $e_1, \dots, e_4$ be the coordinate unit vectors. Define

$$D_4 := \left\{ z \in \mathbb{Z}^4 : \sum_{i=1}^{4} z_i \text{ is even} \right\}, \tag{6.1}$$

$$\mathcal{R} := \{\pm e_i \pm e_j : 1 \le i < j \le 4\}, \tag{6.2}$$

$$P := \{x \in \mathbb{R}^4 : u \cdot x \le 1 \text{ for all } u \in \mathcal{R}\}. \tag{6.3}$$

The gauge

$$g(x) := \max_{u \in \mathcal{R}} u \cdot x \tag{6.4}$$

is the sum of the two largest absolute coordinates of x. In particular, g is a norm, $g(u) = 2$ for $u \in \mathcal{R}$, and $g(x) \le \sqrt{2}\,|x|$. The vertices of P are

$$\{\pm e_i : 1 \le i \le 4\} \ \cup \ \left\{ \tfrac{1}{2}(s_1, s_2, s_3, s_4) : s_i \in \{-1, 1\} \right\}. \tag{6.5}$$

Here is an elementary verification of this convex-hull description. If every coordinate has absolute value at most $1/2$, the point lies in the cube spanned by the half-sign vertices. Otherwise a unique coordinate has absolute value $a > 1/2$, and every other coordinate has absolute value at most $1 - a$. For $a < 1$, the point is $2a - 1$ times the corresponding signed unit vector plus $2 - 2a$ times a point in the half-sign cube. The case $a = 1$ is the signed unit vector itself. Conversely, every displayed generator satisfies the root inequalities. It follows that

$$\overline{B}(0, 1/\sqrt{2}) \subseteq P \subseteq \overline{B}(0, 1). \tag{6.6}$$

The cells $a + P$, $a \in D_4$, form the ordinary Voronoi tiling of D_4. Indeed, the root comparisons are necessary and imply every lattice comparison

$$\delta \cdot x \le \frac{|\delta|^2}{2}, \qquad \delta \in D_4, \quad x \in P. \tag{6.7}$$

It suffices to check the vertices in (6.5). At $\pm e_i$, use $|\delta|^2 \ge 2|\delta_i|$ for an even-sum integer vector; at a half-sign vertex, use $\sum_i |\delta_i| \le \sum_i \delta_i^2$. For the first bound, $\delta_i^2 \ge 2|\delta_i|$ if $|\delta_i| \ge 2$, whereas $|\delta_i| = 1$ requires another odd coordinate. Lattice coercivity gives nearest sites. Root neighbors share genuine facets, since at their midpoint every third lattice site is strictly farther away. Thus the tiling with inballs of radius $1/\sqrt{2}$ is paired.

Lemma 10 (First subdivision)**.** *A standard coarse pair $\big(\overline{B}(a, 3/\sqrt{2}), a + 3P\big)$, with $a \in 3D_4$, can be subdivided into 25 paired tiles of inradius $1/\sqrt{2}$, centered at a and at $a + u$, $u \in \mathcal{R}$. Every complete exterior facet and its contact point remain unchanged. Such subdivisions may be performed independently in any locally finite collection of standard coarse parents in an otherwise arbitrary whole-space paired tiling.*

Proof. Translate a to zero. The affine scores of the new sites are $1/2$ at zero and $2u \cdot x - 3/2$ at u. Their restricted power cells in $3P$ are the central cell P and

$$Q_u := \{x \in \mathbb{R}^4 : 1 \le u \cdot x \le 3, \ u \cdot x \ge v \cdot x \text{ for all } v \in \mathcal{R}\}, \qquad u \in \mathcal{R}. \tag{6.8}$$

The cone inequalities at u need only be imposed for the vertices v adjacent to u in $\operatorname{conv} \mathcal{R}$. These are exactly the roots with $u \cdot v = 1$, and then $v - u$ is again a root. At $u = e_1 + e_2$, the adjacent roots are

$$e_1 \pm e_3, \quad e_1 \pm e_4, \quad e_2 \pm e_3, \quad e_2 \pm e_4.$$

All other cone inequalities follow from these eight: equivalently, the cone is

$$\min(x_1, x_2) \ge \max(|x_3|, |x_4|). \tag{6.9}$$

Signed permutations give the other cases.

Every internal bounding plane of Q_u is therefore a common tangent plane for the indicated equal balls. Its lower plane $u \cdot x = 1$ is the contact plane with the central ball. Its upper plane

$u \cdot x = 3$ is the old parent facet F_u, with unchanged contact point $3u/2$. Every point of this facet satisfies $u \cdot x \geq v \cdot x$ for all $v \in \mathcal{R}$, so Q_u owns it in its entirety. In fact,

$$Q_u \cap \partial(3P) = F_u, \qquad Q_u \cap F_v = F_u \cap F_v. \tag{6.10}$$

For if any $v \cdot x = 3$, the cone inequalities force $u \cdot x = 3$. The entire boundary face complex is thus preserved, including lower-dimensional faces. The restricted power cells tile the parent face-to-face; the boundary identities show that independent subdivisions still fit face-to-face. Their contact points match both unchanged exterior pairs and subdivided neighboring parents.

Every child lies in its parent $3P$, and its center lies within $\sqrt{2}$ of the parent center. Its circumradius about its own center is consequently at most $3+\sqrt{2}$. All its facet normals belong to $\mathcal{R}$. The central child has 24 facets; every other child has at most ten. □

Applying the subdivision to every parent produces exactly the site set

$$A_0 := \{c \in D_4 : k(c) \in \{0, 2\}\}, \qquad k(c) := \#\{i : c_i \not\equiv 0 \pmod 3\}, \tag{6.11}$$

with all radii equal to $1/\sqrt{2}$. Indeed, sites with $k = 0$ are the coarse centers. For $k = 2$, subtract the two signed unit residues to obtain the unique even-sum coarse center in $3D_4$; conversely, every root child has this form. The global A_0 power diagram is therefore paired.

6.2. **A finite interpolation lemma.** For an optional site $c \in D_4 \setminus A_0$, let $\sigma(c) = (\sigma_1(c), \dots, \sigma_4(c))$, where

$$\sigma_i(c) = \begin{cases} 0, & c_i \equiv 0 \pmod 3, \\ 1, & c_i \equiv 1 \pmod 3, \\ -1, & c_i \equiv 2 \pmod 3. \end{cases}$$

Writing $\sigma = \sigma(c)$, define the finite associated set

$$\mathcal{Q}(c) = \begin{cases} \{c - \sigma\}, & k(c) = 1, \\ \{c + \frac{1}{2}\sigma\}, & k(c) = 4, \\ \{c - \sigma,\ c + \frac{1}{2}\sigma + \frac{3}{2}e_i,\ c + \frac{1}{2}\sigma - \frac{3}{2}e_i\}, & k(c) = 3, \end{cases} \tag{6.12}$$

where, in the last case, i is the unique index with $\sigma_i = 0$. These points need not be lattice sites. For $H \subseteq \mathbb{R}^4$, put

$$C(H) = \{c \in D_4 \setminus A_0 : \mathcal{Q}(c) \subseteq H\}. \tag{6.13}$$

Call a local membership rule *orthogonal and read-once* if it is a disjunction of conjunctions of inequalities

$$u_i \cdot x \leq 6n_i,$$

where the u_i are at most four mutually orthogonal roots, $n_i \in \mathbb{Z}$, and each inequality occurs in just one conjunction. The constant rules TRUE and FALSE are allowed.

Lemma 11 (Integer-certified interpolation)**.** *Suppose that, for every optional site c, membership in H on all points of $\mathcal{Q}(b)$, for optional sites $b \in c + [-2, 2]^4$, agrees with some orthogonal read-once rule. The rule may depend on c. Then the balls of radius $1/\sqrt{2}$ centered at $A_0 \cup C(H)$ have a paired whole-space Voronoi diagram. Every new cell has at most 24 facets and circumradius at most $\sqrt{3}$. Every old cell is contained in its original A_0 cell.*

Proof, including reduction to the certificate. For a selected site c, define

$$N_c = \{u \in \mathcal{R} : c + u \in A_0 \cup C(H)\}, \tag{6.14}$$

$$W_c = \{c + y : u \cdot y \leq 1 \text{ for every } u \in N_c\}. \tag{6.15}$$

We must prove that W_c is bounded and satisfies every other-site power comparison. At a doubled vertex $v = 2y$, comparison with $c + \delta$ is equivalent to

$$v \cdot \delta \leq |\delta|^2. \qquad v \cdot \delta > |\delta|^2 \quad \Longleftrightarrow \quad |2\delta - v|^2 < |v|^2. \tag{6.16}$$

The checker in Appendix A enumerates all independent four-tuples of roots, solves $u \cdot v = 2$ by Cramer's rule with integer determinants, and obtains exactly 864 possible doubled

vertices. Every vertex of any bounded root-halfspace intersection occurs on this list. Three independent root equations through zero give, by cofactors and primitive reduction, exactly 48 oriented possible ray directions. These suffice to detect unboundedness: the complete central root arrangement refines any recession cone into pointed cones, and every nonzero such cone contains an arrangement ray. Consequently the recession cone is zero precisely when every listed ray violates at least one of its present homogeneous inequalities.

For every optional residue type, first impose the root neighbors already in A_0. At each remaining candidate vertex, enumerate all strict lattice competitors using the completed-square inequality in (6.16). This enumeration is finite and exact; no truncation hypothesis is used. Mark a vertex as bad if a competitor belongs to A_0. For every unmarked candidate the checker proves that

$$|v|^2 \le 12, \qquad |\delta_i| \le 2 \quad (i = 1, \dots, 4) \quad \text{for every strict competitor } \delta. \tag{6.17}$$

Thus all remaining potential obstructions involve only the optional sites in $c + [-2, 2]^4$. The bound $|v|^2 \le 12$ will give circumradius $\sqrt{3}$ once the bad vertices and all recession rays have been excluded.

Translations by $6\mathbb{Z}^4$, independent coordinate sign changes, and coordinate permutations preserve D_4, A_0, the associated-set rule, and the permitted integer-offset planes. Reducing coordinates modulo 6 into $[-3, 3]$, taking absolute values, and sorting therefore reduces c to one of the following eleven representatives:

$$\begin{gathered}(0,0,0,2), \quad (0,0,1,3), \quad (0,1,1,2), \quad (0,2,2,2),\\ (0,2,3,3), \quad (1,1,1,1), \quad (1,1,1,3), \quad (1,1,2,2),\\ (1,2,2,3), \quad (1,3,3,3), \quad (2,2,2,2).\end{gathered} \tag{6.18}$$

For each representative, collect the source, its optional root neighbors, and every strict competitor of every candidate vertex that satisfies all old-site power comparisons. Collect all their doubled associated points $q \in 2\mathcal{Q}(b)$. For each root u, the rule $u \cdot x \le 6n$ becomes the integer test $u \cdot q \le 12n$. The checker enumerates every distinct membership pattern as n ranges over $\mathbb{Z}$, including the empty and full patterns. There are exactly 48 unordered signed orthogonal root bases. For each basis, take every choice of its four one-plane patterns and every set partition of the four planes. There are 15 partitions, giving disjunctions of disjoint conjunctions. Empty or full plane patterns pad rules using fewer than four planes. An orthogonal root set of size at most four extends to a root basis: roots orthogonal to a given root either use its two coordinates with opposite relative signs or use its complementary coordinate pair. Thus this list covers the hypothesis exactly.

Convert each associated-point pattern to site selection by requiring all points of $\mathcal{Q}(b)$. Whenever the source is selected, the checker verifies that every recession ray is blocked by a selected root neighbor, every old-bad vertex is blocked, and every remaining feasible vertex has no selected strict competitor. The final verified totals are 8,870 distinct associated-point patterns and 2,907 patterns with selected source. The complete program uses only Python's standard library and integer arithmetic, recomputes its geometric catalog, imports no search data, and exits successfully. Its per-source counts and execution command are in Appendix A.

It follows that W_c is bounded, is contained in $\overline{B}(c, \sqrt{3})$, and satisfies every selected-site power comparison. Outside W_c, one of its selected root neighbors has strictly smaller power. Hence W_c is exactly the new site's Voronoi cell.

The full site set lies in D_4, so its equal balls pack. It contains the periodic Delone set A_0, so nearest sites exist and all cells are bounded and locally finite. Every new facet is a root-contact facet. Old cells only shrink; a new facet between an old and a new cell is already a facet of the new cell, and an old-old noncontact facet cannot be created by shrinking. Thus every facet is a contact facet. At the midpoint of a tangent equal-radius pair, every third lattice site has strictly larger power, so each contact gives an actual facet. The ordinary Voronoi face-to-face property finishes the proof. □

The computational statement in Lemma 11 is deliberately explicit. Its exhaustive finite check is a component of this proof, rather than a numerical experiment or an unproved imported geometric assertion.

6.3. **Uniform support, a standard core, and localization.** For a set $H \subseteq \mathbb{R}^4$ and $t \geq 0$, write

$$H + tP = \{h + tp : h \in H,\ p \in P\}, \tag{6.19}$$

$$H \ominus tP = \{x \in \mathbb{R}^4 : x + tP \subseteq H\}. \tag{6.20}$$

Neither operation requires H to be convex.

Lemma 12 (Core, support, and localization)**.** *Under the hypotheses of Lemma 11, every standard D_4 cell meeting $H \ominus 5P$ is present as a complete standard cell. All changes from the A_0 diagram are contained in $H + 5P$. If every standard coarse parent $a + 3P$ meeting $H + 5P$ is available in a current whole-space paired tiling, one may first subdivide exactly these parents and then insert $C(H)$. The resulting whole-space paired tiling changes only those parents, all of which lie in $H + 11P$. It has the stated standard core. Every ball in the resulting affected pairs has radius $1/\sqrt{2}$, and its cell has at most 24 facets and shape ratio at most $3\sqrt{2} + 2$.*

Proof. Directly from (6.12), we have $g(q - c) \leq 2$ for every $q \in \mathcal{Q}(c)$. Take $x \in H \ominus 5P$ and a standard cell $a + P$ containing x. For every optional root neighbor $b = a + u$ and every $q \in \mathcal{Q}(b)$,

$$g(q - x) \leq g(q - b) + g(u) + g(a - x) \leq 2 + 2 + 1 = 5. \tag{6.21}$$

Thus all optional root neighbors of a are selected; the nonoptional ones belong to A_0. The site a itself is selected by the same estimate with $u = 0$, if it is optional, and otherwise belongs to A_0. All roots are present, so a has its complete standard cell: the root inequalities cut it down to $a + P$, and no selected lattice site can cut $a + P$ further, by the full-lattice comparison proved in Section 6.1. This proves the core claim, including cells incident to the core boundary.

For any selected new site c, choose $q \in \mathcal{Q}(c) \subseteq H$. Since its new cell lies in $\overline{B}(c, \sqrt{3})$ and $g(y) \leq \sqrt{2}|y|$,

$$W_c \subseteq q + (2 + \sqrt{6})P \subseteq H + (2 + \sqrt{6})P \subseteq H + 5P. \tag{6.22}$$

Adding sites changes old cells only on the union of new cells. Hence this also bounds the support of the insertion.

Select all standard coarse parents meeting $H + 5P$. If a selected parent meets that set at z, its entire parent lies in $z + 6P$, hence in $H + 11P$. Subdivide these parents by Lemma 10. Each proposed new cell from the global $A_0 \cup C(H)$ diagram lies in their union. At any point x of that cell, the current subdivided tiling has a fine A_0 owner from one of these parents. By Lemma 3, its power is the actual current minimum. The global A_0 minimum is no larger than this owner's power. The verified new power is no larger than that global minimum, and therefore no larger than the actual current minimum.

Every required old root neighbor $b = c + u$ is actually present. Indeed, its contact midpoint with c belongs to W_c and lies in the interior of b's global A_0 cell: among A_0 sites, b alone minimizes power there. Thus b's coarse parent meets $H + 5P$ and was selected. New root neighbors are inserted simultaneously. Outside W_c, one of these actual root neighbors has smaller power. Therefore W_c is still exactly the new power cell in the actual surrounding tiling.

Packing is also preserved: every new closed ball lies in its W_c and hence in the selected parents' union. The old balls supplied by those parents and the new balls have distinct centers in D_4, whose minimum center separation is $\sqrt{2}$. Old ball interiors outside the selected union are disjoint from its interior. The old minimum power is attained by Lemma 3, and the new equal-radius lattice powers tend to infinity with their centers. Hence the enlarged minimum is attained at every point. Local finiteness follows because old cells shrink and new cells have uniformly bounded diameter and lattice centers. The same facet argument as in Lemma 11

proves pairing. Unselected parents and all other complete pairs remain unchanged. New cells are contained in the selected parents' union, although a new cell may cross an internal parent facet.

The complete standard core also survives this localization. For $x \in H \ominus 5P$ and $x \in a + P$, the whole cell $a + P$ lies in $x + 2P \subseteq H$. Its A_0 parent, if a is an old fine site, was selected because its global A_0 cell contains x. Its actual old power comparisons throughout $a + P$ follow from the same minimum-power argument above; all new-site comparisons follow from $a + P$ being a full-lattice Voronoi cell. Every required old root neighbor is present, since its contact point lies in $H + 5P$ and in its A_0 parent. Therefore the actual cell at a is exactly $a + P$, including when $a \in A_0$.

An old subdivided child shrinks from a cell of circumradius at most $3 + \sqrt{2}$. All its old facet normals, including exterior-parent facets, are roots. Newly added facets also have root normals. Because each plane is tangent to the same fine inball, there is at most one facet for each of the 24 root directions. New cells have circumradius at most $\sqrt{3}$. These bounds yield the claimed uniform ratio. Scaling all coordinates by h scales all lengths in this lemma by h. □

6.4. **Admissible neighborhoods of an arbitrary closed set.** Fix the orthogonal roots

$$u_1 = e_1 + e_2, \qquad u_2 = e_1 - e_2, \qquad u_3 = e_3 + e_4, \qquad u_4 = e_3 - e_4, \tag{6.23}$$

and write $y_i(x) = u_i \cdot x$. These coordinates satisfy

$$\max_{1 \le i \le 4} |y_i(x)| \le g(x) \le 2 \max_{1 \le i \le 4} |y_i(x)|. \tag{6.24}$$

The lower bound follows from the corresponding root inequalities. For the upper bound, the identities $x_1 = \frac{y_1+y_2}{2}$ and $x_2 = \frac{y_1-y_2}{2}$, together with the analogous identities for x_3, x_4, bound each $|x_i|$ by $\max_j |y_j|$.

Let $F \subset \mathbb{R}^4$ be nonempty and closed, and abbreviate $F_t = F + tP$. Fix

$$M = 1008, \qquad L = 10000. \tag{6.25}$$

For $h > 0$ and $z \in \mathbb{Z}^4$, define

$$Q_z = \{x \in \mathbb{R}^4 : Mhz_i \le y_i(x) \le Mh(z_i + 1) \text{ for } i = 1, \dots, 4\}. \tag{6.26}$$

Assign the sixteen parity vectors of z distinct labels $j(z) \in \{1, \dots, 16\}$, and set $t_z = 18 + 18j(z)$. Thus $36 \le t_z \le 306$. Expand the boxes by setting

$$B_z = \{x \in \mathbb{R}^4 : Mhz_i - t_z h \le y_i(x) \le Mh(z_i + 1) + t_z h \text{ for } i = 1, \dots, 4\}. \tag{6.27}$$

Select precisely those z for which $Q_z \cap F_{Lh} \neq \varnothing$, and let H_h be the union of their boxes B_z. The full box family is locally finite, including when F is unbounded, so H_h is closed.

Lemma 13 (Admissible neighborhoods)**.** *The sets H_h satisfy the scaled hypothesis of Lemma 11 and the uniform bounds*

$$F_{10000h} \subseteq \operatorname{int}(H_h), \qquad H_h \subseteq F_{13240h}. \tag{6.28}$$

Proof. We first prove the bounds. Every point of F_{Lh} lies in a selected Q_z and strictly inside its expansion B_z. If $x \in B_z$ for a selected z, choose $q \in Q_z \cap F_{Lh}$. Both points belong to B_z, so $|y_i(x - q)| \le (M + 612)h$ for every i. By (6.24),

$$g(x - q) \le 2(M + 612)h = 3240h. \tag{6.29}$$

Consequently $x \in F_{(L+3240)h}$. This proves (6.28) without using convexity or boundedness of F.

We next verify the local membership rule. The associated points of all optional sites $b \in c + [-2h, 2h]^4$ in the scaled interpolation hypothesis lie in

$$W = c + \left[-\frac{7h}{2}, \frac{7h}{2}\right]^4. \tag{6.30}$$

The projection of W on each y-coordinate has length $14h$. Discard any selected box with a side inequality that is constantly false on W. For every remaining box, each of its four coordinate intervals meets the corresponding projection of W; intersection of the whole box with W is not required.

Any two remaining boxes differ by at most one in each grid index. Indeed, indices separated by two give coordinate intervals separated by at least

$$(M - 612)h = 396h > 14h. \tag{6.31}$$

Distinct remaining boxes therefore have distinct parity colors and distinct values of t_z. In any fixed coordinate direction, distinct side planes from these boxes have offsets separated by at least $18h$. With the same grid base and orientation this follows from the distinct colors; with opposite orientations at the same base the separation is at least $72h$; with different bases it is at least $396h$. Coincident planes from distinct boxes would require equal colors, which have already been excluded.

It follows that at most one side plane in each of the four directions is nonconstant on W. Each such plane belongs to just one remaining box. Constantly true inequalities may be omitted. A box whose inequalities are all constantly true makes the union rule constantly true. Otherwise, membership in each remaining box is the conjunction of its nonconstant inequalities, and membership in their union is the disjunction of these conjunctions. There are at most four nonconstant planes, their normals are mutually orthogonal, and no plane is reused. Every plane offset is a multiple of $6h$, because M and all t_z are divisible by 6. This is precisely the scaled hypothesis of Lemma 11. □

6.5. **Nested replacements and stabilization.**

Lemma 14 (Nesting)**.** *The entire replacement support at scale $\frac{h}{3}$ lies in the standard core produced at scale h:*

$$H_{h/3} + \frac{11h}{3}P \subseteq H_h \ominus 5hP. \tag{6.32}$$

Proof. By Lemma 13, the left-hand side of (6.32) is contained in $F_{13251h/3}$. Since

$$\frac{13251}{3} = 4417 < 9995, \tag{6.33}$$

it is contained in the interior of F_{9995h}. If $x \in F_{9995h}$, convexity of P gives

$$x + 5hP \subseteq F_{10000h} \subseteq H_h. \tag{6.34}$$

Thus the entire next support in fact lies in the interior of the standard core. Only P is used as a convex set here. □

Proof of Theorem 1 in dimension four. Put $F = \mathbb{R}^4 \setminus \Omega$. If F is empty, a sufficiently small standard D_4 tiling proves the theorem. Otherwise choose $h_0 > 0$ with $\frac{3h_0}{\sqrt{2}} \le \varepsilon$, set $h_n = h_0 3^{-n}$, and start with the standard coarse tiling whose cells are $a + 3h_0P$, for $a \in 3h_0D_4$.

At stage $n \ge 0$, apply the scaled replacement of Lemma 12 to H_{h_n}. At stage zero all required coarse parents are available. Inductively, the next operation has coarse scale $3h_{n+1} = h_n$, and Lemma 14 places its entire support, including all selected parents, in the preceding standard core. Every required parent is therefore an already present complete standard cell of the correct scale. Each stage gives a legitimate whole-space paired tiling. This also covers an infinite collection of selected parents: at each fixed positive scale the collection is locally finite, and the local replacements preserve its exterior facets, as in Lemmas 10 and 12.

The support S_n of all changes at stage n satisfies

$$S_n \subseteq H_{h_n} + 11h_nP \subseteq F_{13251h_n}. \tag{6.35}$$

Every compact set $K \subset \Omega$ has positive g-distance from F. Consequently all sufficiently late supports avoid an open neighborhood of K. From that stage onward, every complete pair meeting this neighborhood is unchanged: the operations replace only selected whole parents,

and those parents are contained in the supports. Thus stabilization concerns whole pairs, rather than only their intersections with K.

Define the final family to consist of all complete ball–tile pairs occurring at some finite stage and unchanged at every later stage. Every point of Ω lies in such a pair by the preceding stabilization argument.

No surviving tile meets F. The inclusion

$$F_{9995h_n} \subseteq H_{h_n} \ominus 5h_nP \tag{6.36}$$

places an open neighborhood of F in the standard core. Every tile incident to a point $p \in F$ is therefore a standard h_nD_4 cell, even if it touches p only at a vertex: otherwise its nearby interior would overlap one of the complete standard cells covering that neighborhood. At the next stage this cell is a selected coarse parent. Indeed, it contains p, while

$$p \in H_{h_{n+1}} \subseteq H_{h_{n+1}} + 5h_{n+1}P. \tag{6.37}$$

It is replaced by balls of strictly smaller radii and cannot survive. A permanently surviving pair touching F would contradict this at any stage after its stabilization. The final union is therefore exactly Ω, and every final tile is contained in Ω.

The whole-pair stabilization lemma, Lemma 5, now gives a countably infinite paired tiling of exactly Ω and identifies its tangency polytopes.

All surviving radii are positive finite-stage radii and are at most $\frac{3h_0}{\sqrt{2}} \leq \varepsilon$. Lemma 12 gives at most 24 facets and shape ratio at most $3\sqrt{2}+2$ at every finite stage, and hence in the final family. □

7. Scope and provenance

The planar and three-dimensional local replacements and the four-dimensional integer-certified interpolation are different constructions. The common lemmas do not assert a dimension-reduction or a refinement theorem for arbitrary lattices. In particular, the admissible-neighborhood rule used in dimension four is retained explicitly.

This edition consolidates the three AI-generated source notes *Circle Packings of Prescribed Planar Open Sets*, *Sphere Packings Whose Tangency Polytopes Tile a Prescribed Open Subset of* $\mathbb{R}^3$, and *Sphere Packings Whose Tangency Polytopes Tile Prescribed Open Subsets of* $\mathbb{R}^4$. Common setup and stabilization arguments are stated once. Historical self-audit appendices are omitted. This is an unrefereed research note.

References

[BG25] Ahmed Bou-Rabee and Ewain Gwynne. *Random walk on sphere packings and Delaunay triangulations in arbitrary dimension.* Proceedings of the London Mathematical Society **131** (2025), e70084. arXiv:2405.11673v2.

Appendix A. Complete exact checker

The following source is included in full and also embedded as a file attachment in this PDF. Save it as `verify_d4.py` and run

```
python3 verify_d4.py
```

with no optimization flag. It uses only the standard library. All determinant, sign, distance, and Boolean tests use exact integers.

The execution checked 864 doubled arrangement vertices, 48 oriented rays, 48 orthogonal root bases, and 15 set partitions. In the order of the eleven source representatives in Section 6.2, the numbers of distinct associated-point patterns are

$$460,\ 460,\ 1404,\ 1421,\ 460,\ 460,\ 1421,\ 460,\ 1404,\ 460,\ 460.$$

They sum to 8870. Of these, 2907 select the source. All boundedness and power comparisons passed.

The SHA-256 digest of the included checker, split across two lines, is

653a9029888607e74dd313ce54764808d9
fab6dc1cdde1d16d300cb57ccede04

```
"""Independent integer proof checker for orthogonal read-once D4 root cutoffs.

No imports from search/generation code or third-party packages.  The accompanying
note proves reduction to eleven source residues and48 orthogonal root bases.
This recomputes the arrangement, its local power tests, and every finite case.
"""
from itertools import product, combinations, combinations_with_replacement, permutations
from functools import lru_cache
from math import gcd, isqrt

def dot(a, b):
    return sum((x * y for x, y in zip(a, b)))

def add(a, b):
    return tuple((x + y for x, y in zip(a, b)))

def det(rows):
    n = len(rows)
    ans = 0
    for p in permutations(range(n)):
        v = (-1) ** sum((p[i] > p[j] for i in range(n) for j in range(i + 1, n)))
        for i in range(n):
            v *= rows[i][p[i]]
        ans += v
    return ans
R = []
for i, j in combinations(range(4), 2):
    for a, b in product((-1, 1), repeat=2):
        u = [0] * 4
        u[i] = a
        u[j] = b
        R.append(tuple(u))
V = set()
W = set()
for rows in combinations(R, 4):
    D = det(rows)
    if not D:
        continue
    v = []
    for j in range(4):
        M = [list(r) for r in rows]
        for r in M:
            r[j] = 1
        num = 2 * det(M)
        assert num % D == 0
        v.append(num // D)
    V.add(tuple(v))
for rows in combinations(R, 3):
    w = tuple(
        (-1) ** j * det([[r[k] for k in range(4) if k != j] for r in rows])
        for j in range(4)
    )
    g = gcd(*w)
    if g:
        w = tuple((x // g for x in w))
        W.add(w)
        W.add(tuple((-x for x in w)))
assert len(V) == 864 and len(W) == 48
V = sorted(V)
W = sorted(W)
vmask = {v: sum((1 << i for i, u in enumerate(R) if dot(u, v) > 2)) for v in V}
wmask = {w: sum((1 << i for i, u in enumerate(R) if dot(u, w) > 0)) for w in W}
```

```
violators = {}
for v in V:
    q = isqrt(dot(v, v)) + 1
    ranges = [range((x - q) // 2, (x + q) // 2 + 2) for x in v]
    violators[v] = [
        d for d in product(*ranges)
        if sum(d) % 2 == 0 and dot(v, d) > dot(d, d)
    ]
assert all((vmask[v] or not violators[v] for v in V))
assert all(wmask.values())

def old(c):
    return sum((x % 3 != 0 for x in c)) in (0, 2)

def typ(c):
    return tuple((x % 3 for x in c))

@lru_cache(None)
def local_catalog(r):
    c = list(r)
    if sum(c) % 2:
        c[0] += 3
    c = tuple(c)
    mask = sum((1 << i for i, u in enumerate(R) if old(add(c, u))))
    bad = []
    possible = []
    for v in V:
        if vmask[v] & mask:
            continue
        if any((old(add(c, d)) for d in violators[v])):
            bad.append(vmask[v])
        else:
            assert dot(v, v) <= 12
            assert all((abs(x) <= 2 for d in violators[v] for x in d))
            possible.append(v)
    return (bad, possible)
for r in product(range(3), repeat=4):
    if not old(r):
        local_catalog(r)
print('Certified locality: every old-valid candidate vertex has |v|\u00b2\u226412 '
      'and all possible new violators have |delta_i|\u22642.', flush=True)

@lru_cache(None)
def holes(c):
    s = tuple((0 if x % 3 == 0 else 1 if x % 3 == 1 else -1 for x in c))
    k = sum((x != 0 for x in s))
    if k == 1:
        return (tuple((2 * (x - z) for x, z in zip(c, s))),)
    if k == 4:
        return (tuple((2 * x + z for x, z in zip(c, s))),)
    assert k == 3
    i = s.index(0)
    return (tuple((2 * (x - z) for x, z in zip(c, s))),) + tuple(
        tuple(
            2 * x + z + (a if j == i else 0)
            for j, (x, z) in enumerate(zip(c, s))
        )
        for a in (-3, 3)
    )

def setup(c):
    bad, possible = local_catalog(typ(c))
    sites = {c}
    sites.update((add(c, u) for u in R if not old(add(c, u))))
    for v in possible:
```

```
        sites.update((add(c, d) for d in violators[v]))
    assert all((not old(a) for a in sites))
    sites = sorted(sites)
    ids = {a: i for i, a in enumerate(sites)}
    qs = sorted({q for a in sites for q in holes(a)})
    qid = {q: i for i, q in enumerate(qs)}
    needs = [sum((1 << qid[q] for q in holes(a))) for a in sites]
    oldmask = sum((1 << i for i, u in enumerate(R) if old(add(c, u))))
    neighbors = [(i, ids[add(c, u)]) for i, u in enumerate(R) if not old(add(c, u))]
    rules = [
        (vmask[v], sum((1 << ids[add(c, d)] for d in violators[v])))
        for v in possible
    ]
    planes = []
    for r in R:
        heights = [dot(r, q) for q in qs]
        options = {0}
        for n in range(min(heights) // 12, max(heights) // 12 + 2):
            options.add(sum((1 << i for i, h in enumerate(heights) if h <= 12 * n)))
        planes.append(sorted(options))
    return (ids[c], needs, oldmask, neighbors, rules, planes, bad)
bases = [
    ids for ids in combinations(range(24), 4)
    if all((dot(R[i], R[j]) == 0 for i, j in combinations(ids, 2)))
]
assert len(bases) == 48
representatives = [
    c for c in combinations_with_replacement(range(4), 4)
    if sum(c) % 2 == 0 and (not old(c))
]
assert len(representatives) == 11
total = selected_patterns = 0
for c in representatives:
    source, needs, oldmask, neighbors, rules, planes, bad = setup(c)
    seen = set()
    for ids in bases:
        for masks in product(*(planes[i] for i in ids)):
            A, B, C, D = masks
            patterns = [
                A | B | C | D, A & B | C & D, A & C | B & D,
                A & D | B & C, A & B & C & D
            ]
            for i in range(4):
                rest = [masks[j] for j in range(4) if j != i]
                patterns.append(masks[i] | rest[0] & rest[1] & rest[2])
            for i, j in combinations(range(4), 2):
                rest = [masks[k] for k in range(4) if k not in (i, j)]
                patterns.append(masks[i] & masks[j] | rest[0] | rest[1])
            assert len(patterns) == 15
            for H in patterns:
                if H in seen:
                    continue
                seen.add(H)
                total += 1
                if H & needs[source] != needs[source]:
                    continue
                selected_patterns += 1
                present = sum(
                    1 << i for i, need in enumerate(needs) if H & need == need
                )
                mask = oldmask + sum((1 << i for i, j in neighbors if present & 1 << j))
                assert all((m & mask for m in wmask.values())), ('unbounded', c)
                assert all((m & mask for m in bad)), ('old-power', c)
                assert all(
                    m & mask or not present & competitors
```

```
                    for m, competitors in rules
                ), ('new-power', c)
    print('VALID source', c, 'hole patterns', len(seen), flush=True)
assert total == 8870
print(
    'PASS:', total, 'hole patterns;', selected_patterns,
    'selected-source patterns; all48 orthogonal root bases and15 set partitions.',
    flush=True
)
print(
    'All radii remain1/sqrt(2), at most24 facets, '
    'and all new circumradii are at most sqrt(3).',
    flush=True
)
```